Who Hunts Who?

Comparing 4 Knowledge Graph Construction Methods for Predator–Prey Relations

Bek Zoltan, Lohse Armin, Reil Nino Joshua, Wallnöfer Andreas

Natural Language Processing

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Motivation & Task

- Goal: automatically build a predator–prey knowledge graph from free text.
- Source: Wikipedia pages for 18 animals across three trophic levels.
- No animal-specific hand-written rules — only general NLP signals.
- Central question: which extraction approach yields the best trade-off between precision and recall?

Three trophic levels

Top predators (6)
Hawk, Eagle, Barn Owl, Fox, Wolf, Weasel

Mid-level (4)
Salamander, Lizard, Spider, Bat

Prey (8)
Mouse, Rabbit, Squirrel, Grasshopper, Cricket, Caterpillar, Fly, Beetle

Pipeline Overview

Wikipedia corpus (18 pages) → Sentence preprocessing → Extraction Methods A–D → Knowledge graph

Corpus

Pages fetched via `wikipedia-api`, cached as JSON. Text is stripped of markup, parentheticals, and citation markers, then split into sentences.

Gold standard

38 manually curated predator→prey edges. The same benchmark is used for all four methods to ensure a fair comparison.

Metrics

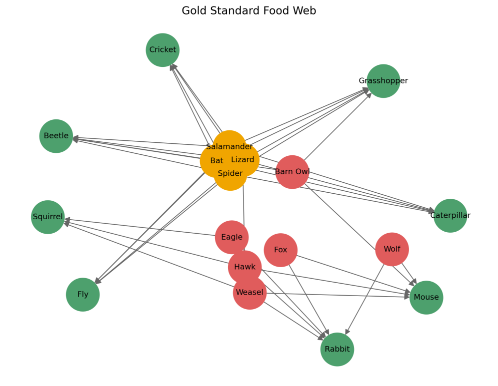
Edge-level Precision, Recall, and F1 score. Only edges within valid trophic directions are considered.

Gold Standard

Construction rules

- Manually constructed from Wikipedia text.
- Directed edges: predator $\rightarrow$ prey.
- Trophic constraints enforced:
  - top predators may target mid-level *and* prey.
  - mid-level predators may target prey only.
- 38 edges in total across 10 source animals.

Gold standard graph



Method A — Co-occurrence

How it works

1. For each source animal, iterate over sentences in its Wikipedia page.
2. If any target animal is mentioned in the same sentence: add edge source $\rightarrow$ target.
3. No verb, no syntactic analysis — pure proximity.

Example sentence:
"Foxes and rabbits both live in the forest."

$\Rightarrow$ adds Fox$\rightarrow$Rabbit
even though the sentence is about their *habitat*.

Design trade-offs

Strength: highest possible recall — every real edge that co-occurs in a sentence will be found.

Limitation: no predation signal is required. Any two animals appearing together (as competitors, taxonomic relatives, or examples) become an edge.

Limitation: no direction inference — edges are always assigned as source $\rightarrow$ target based on which page is being read.

Expected: very high recall, low precision.

Method A vs Gold Standard

Gold standard

Gold Standard Food Web

Method A

Method A - Co-occurrence

Scores: Precision $= 0.346$, Recall $= 1.000$, F1 $= 0.514$ (TP $= 36$, FP $= 68$, FN $= 0$)

Method B — Predation Verb Patterns

How it works

1. For each sentence containing the source animal, extract a window around both animals.
2. Requires at least one predation verb or phrase inside that window:
3. Infer direction:
   - Active: “Fox *hunts* Rabbit” → Fox→Rabbit.
   - Passive: “Rabbit *is hunted by* Fox”
     Fox→Rabbit.
4. Category expansion fallback: “eats insects”, expands to “eats fly“

Design trade-offs

Strength: the verb requirement dramatically reduces false positives compared to Method A.

Limitation: paraphrased predation (*“diet consists of. . . ”*, *“food source for. . . ”*) is missed, lowering recall.

Limitation: direction inference from text position is heuristic; complex sentence structures may be mis-directed.

Limitation: plural forms (mice, flies) not handled uniformly in the window search.

Expected: highest precision, moderate recall.

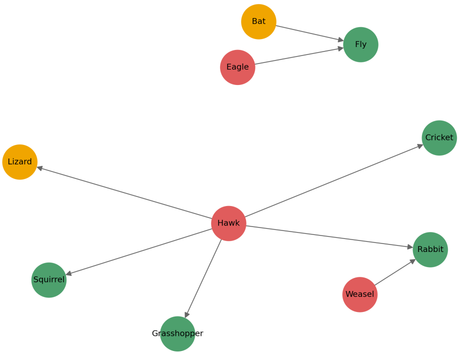
Method B vs Gold Standard

Gold standard

Gold Standard Food Web

Method B

Method B - Dependency Patterns



Scores: Precision $= 0.625$, Recall $= 0.139$, F1 $= 0.227$ (TP $= 5$, FP $= 3$, FN $= 31$)

Method C — OpenIE-style Relation Extraction

Core idea: extract `(subject, relation, object)` triples.

Step-by-step

1. Build regex from the full predation phrase + keyword vocabulary.
2. Search for `[source] ...[rel] ...[target]` up to 180 characters apart.
3. Handle passive: `[target] ...is eaten by ...[source]`.
4. Category expansion fallback: “eats insects”, expands to “eats fly“

Strength: variable-distance regex captures paraphrases missed by Method B’s fixed window.

Strength: category expansion recovers edges when Wikipedia mentions a category (“insects”) rather than a specific animal.

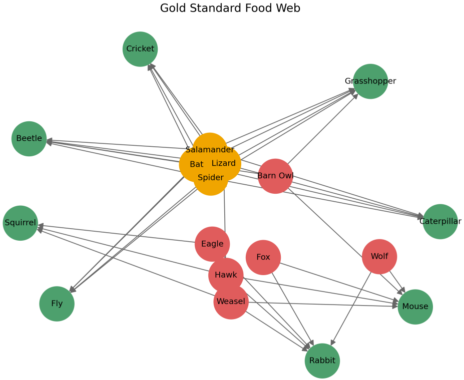
Limitation: generous distance windows may allow subject–relation–object triples that are not actually related.

Limitation: unconstrained category expansion can introduce false positives.

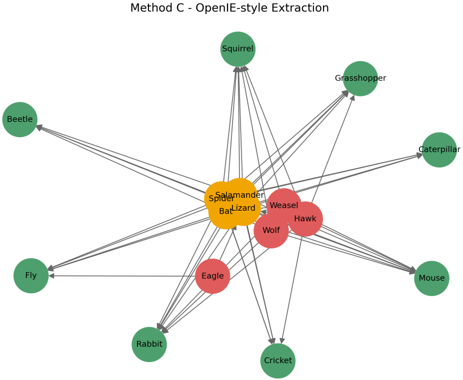
Expected: better recall than B, moderate precision.

Method C vs Gold Standard

Gold standard



Method C



Scores: Precision $= 0.604$, Recall $= 0.806$, F1 $= 0.690$ (TP $= 29$, FP $= 19$, FN $= 7$)

Method D — Hybrid Extraction (Overview)

Method D combines rule-based extraction with constrained fallback expansion:

- **Step 1: Explicit relation extraction**
  - Predicate-driven extraction using verb cues + proximity constraints
- **Step 2: Restricted category expansion**
  - Applied only if Step 1 fails
  - Uses insect-based semantic expansion under strict constraints

Key idea: maximize precision first, then recover recall via controlled fallback.

Method D — Step 1: Explicit Relation Extraction

Goal: extract high-precision predator $\rightarrow$ prey edges

Procedure:

- Identify predation verbs (eat, hunt, kill, consume, attack, etc.)
- Reject sentences containing negation cues (e.g. not, rarely, never)
- Detect entity mentions with lexical variants (e.g. mouse/mice)

Edge creation conditions:

- Both predator and prey appear in the same sentence
- Both are within a token distance threshold to predation verbs:
  - predator distance $\leq d_{pred}$
  - prey distance $\leq d_{prey}$
- OR passive construction detected (“X is eaten by Y”)

Effect: high precision extraction with strict linguistic constraints

Method D — Step 2: Restricted Category Expansion

Goal: recover missed edges from Step 1 under controlled conditions

Trigger condition:

- Only executed if Step 1 does not extract any relation
- Sentence must contain at least one predation cue

Expansion strategy:

- Expand prey candidates using insect taxonomy only
- Apply only to insectivorous source species:
  - Salamander, Lizard, Spider, Bat, Bird

Effect:

- Improves recall without significantly increasing false positives
- Acts as a safety-constrained fallback mechanism

Method D — Strengths & Limitations

Strength

- Combines explicit linguistic rules with controlled fallback expansion
- Dual proximity constraint (predator + prey) improves precision
- Handles passive constructions (e.g. “X is eaten by Y”)
- Lexical variant matching improves robustness (mouse/mice, etc.)
- Negation filtering reduces false positive relations

Limitation

- Relies on manually defined predation vocabulary
- Token-based distance is only a proxy for syntactic structure
- Category expansion is restricted to insect prey
- Cannot capture cross-sentence relations
- Performance depends on sentence-level parsing quality

Overall: best-performing hybrid trade-off between precision and recall

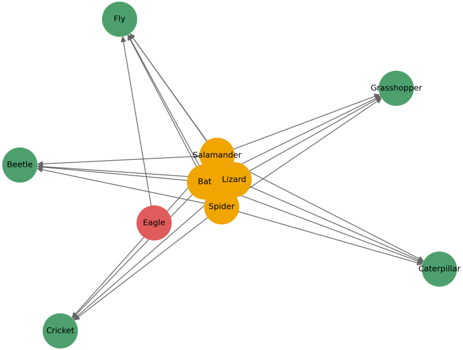
Method D vs Gold Standard

Gold standard

Gold Standard Food Web

Method D

Method D - Hybrid Extraction



Scores: Precision $= 0.889$, Recall $= 0.667$, F1 $= 0.762$ (TP $= 24$, FP $= 3$, FN $= 12$)

Method Design Comparison

	A Co-occurrence	**B** Verb patterns	**C** OpenIE	**D** Hybrid
Predation verb required	No	Yes	Yes	Yes
Direction inference	No	Yes	Yes	Yes
Passive voice handled	—	Yes	Yes	Yes
Category expansion	None	None	Full	Restricted
Proximity constraint	None	Window	Char. distance	Dual token distance
Noisy-context filter	No	No	Yes	Yes
Variant matching	Yes	Partial	Yes	Yes
Expected precision	Low	High	Medium	High
Expected recall	High	Medium	High	Medium-high

Evaluation Metrics

Edge-level evaluation

Let $\hat{E}$ be the set of predicted edges and E^* the gold standard.

$$\text{Precision} = \frac{|\hat{E} \cap E^*|}{|\hat{E}|} \quad \text{Recall} = \frac{|\hat{E} \cap E^*|}{|E^*|}$$

$$F_1 = \frac{2 \cdot P \cdot R}{P + R}$$

- $|\hat{E} \cap E^*|$ = true positives (TP)
- $|\hat{E}| -$ TP = false positives (FP)
- $|E^*| -$ TP = false negatives (FN)

Scope constraint

Only edges that satisfy the trophic direction are considered valid:

- Source must be in top predators $\cup$ mid-level.
- Target must be in mid-level $\cup$ prey.
- Self-loops are never valid.
- Top predators cannot appear as prey targets.

All four methods use this same constraint, implemented via `sanitize_edges()`.

Results

Method	P	R	F1	TP	FP	FN
A — Co-occurrence	0.346	0.100	0.514	36	68	0
B — Dependency Patterns	0.625	0.139	0.227	5	3	31
C — OpenIE-style Relation Extraction	0.604	0.806	0.690	29	19	7
D — Hybrid	0.889	0.667	0.762	24	3	12

Error Analysis

- **Method A — Co-occurrence**
  - High recall due to saturation strategy
  - Very large number of false positives (semantic noise)
  - Lacks directional and relational constraints
- **Method B — Dependency Patterns**
  - Very low recall due to strict pattern dependency
  - Fails on lexical variation and non-canonical syntax
  - High precision but extremely sparse predictions
- **Method C — OpenIE-style**
  - Best recall (0.806) but introduces structural noise
  - Errors from overly permissive relation scopes
  - Sensitive to ambiguous or generic relation phrases
- **Method D — Hybrid**
  - Strong precision (0.889), low FP rate
  - Remaining errors mainly due to:
    - long-distance dependencies beyond threshold
    - missed implicit relations without explicit cue

Discussion

- **Key finding: trade-off between precision and recall**
  - Rule-based constraints significantly improve precision
  - However, stricter filtering reduces recall
- **Best overall method: Hybrid (D)**
  - Achieves best F1 (0.762) across all methods
  - Balances explicit linguistic rules with controlled fallback expansion
  - Strong reduction in false positives compared to C
- **Main bottleneck**
  - Recall limited by proximity thresholds and sentence-level extraction
  - Cross-sentence and long-range dependencies remain unhandled
- **Interpretation**
  - A: recall-heavy baseline
  - B: precision-heavy but brittle system
  - C: recall-maximising noisy extractor
  - D: controlled balance between both extremes

Summary

- **Conclusion:** Combining explicit predation cues with constrained rule-based filtering achieved the best overall extraction performance.

	Method	Core signal	Main characteristic
A	Co-occurrence	Sentence co-occurrence	Highest recall
B	Verb patterns	Predation verb patterns	Explicit linguistic evidence
C	OpenIE-style	Relation extraction + expansion	Balanced recall
D	Hybrid	Verb proximity + restricted expansion	Highest Precision & F1

Method D achieved the highest Precision (0.889) and F1-score (0.762), while Method A achieved perfect Recall through exhaustive edge generation.